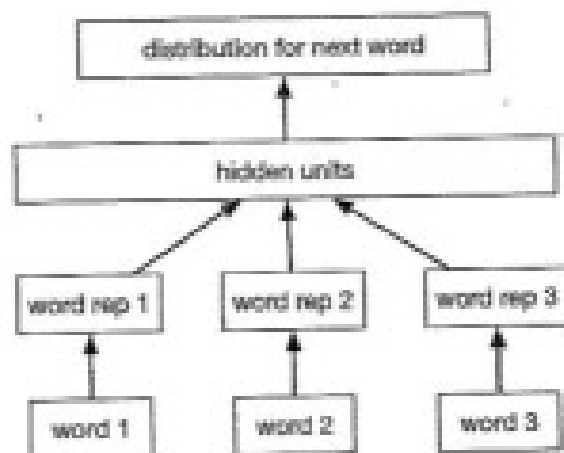


1. In autoregressive language modeling, when we generate from the model (i.e. compute samples from its distribution over sentences), the outputs feed back in to the network as inputs. What is teacher forcing, and why is it used? [8 pts]

During training, we feed in next tokens from training data instead of the model output s.t. it learns correct dependencies (this is teacher forcing). It's used for model to learn correct representations & if its output is wrong for several tokens, the rest of the model



Why is it that the learned word representations for *teach* and *professor* are likely to be far apart? (It is not sufficient to say they are semantically dissimilar. Your answer should mention how the representations are used in the prediction task.) [8 pts]

3. Recall the CycleGAN model for style transfer. ^{from consistency term $1 \rightarrow 2 \rightarrow 1$ & $2 \rightarrow 1 \rightarrow 2$ shouldn't be} very different start & end

(a) If the model is successful, what will it be able to do? [5 pts]

If we give the model an image and ask it to transfer it to another style, it'll successfully return objects but not the style (eg. Van Gogh $\leftrightarrow$ photorealistic horses $\rightarrow$ horses)

(b) What data is required to train the model? [5 pts]

Unmatched images of different styles (eg. Van Gogh paintings & landscape photos), where we specify the styles, don't match images from different styles to each other

(c) What might go wrong if we eliminate the discriminator terms from the cost function? [5 pts]

The model won't be good at generating realistic images in a specific style (it's a GAN, it needs this term). Performance is dependent on the competition between the generator & discriminator.

(d) What might go wrong if we eliminate the reconstruction term from the cost function? [5 pts]

4. In this course, we've discussed several cases where you want to use backprop to compute the gradient of some function with respect to the pixels of an image. Describe two of these cases. For each one, explain what function we compute the gradient of and what the gradient is useful for. (A one-sentence verbal description is sufficient; you don't need to write equations.)

(a) Case 1 [5 pts]:

Grad of loss fct wrt input (image pixels)

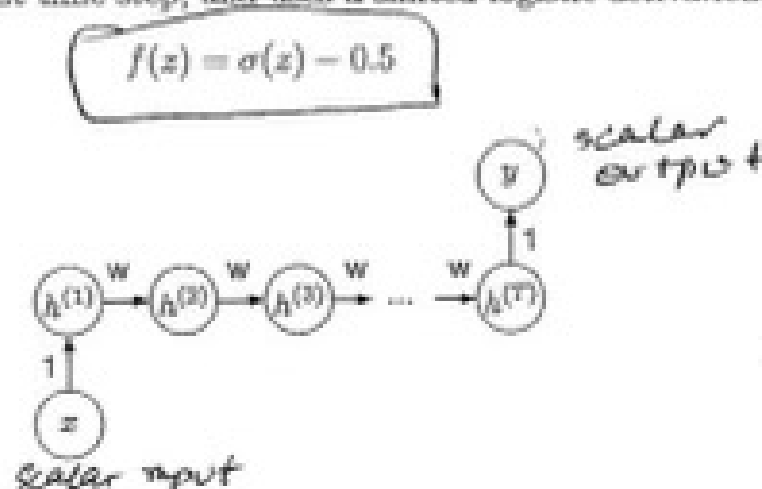
↓
to know which input pixels affect the output the most

(b) Case 2 [5 pts]:

Grad. of a specific unit's activation wrt input

↓
to know which pixels make the unit (→ to refer what the unit represents)

5. Consider the following RNN, which has a scalar input at the first time step, makes a scalar prediction at the last time step, and uses a shifted logistic activation function:



- (a) Write the formula for the derivative $\overline{h}_t$ as a function of $\overline{h}_{t+1}$, for $t < T$. (Use z_t to denote the input to the activation function at time t . You may write your answer in terms of σ' , i.e., you don't need to explicitly write out the derivative of σ .) [8 pts]

$$\begin{aligned}\overline{h}_t &= w \overline{z}^{(t+1)} = w G'(z^{(t+1)}) \overline{h}^{(t+1)} = \\ &= w G'(w h^{(t)}) \overline{h}^{(t+1)}\end{aligned}$$

derivation on
prev. page

- (b) Suppose the input to the network is $x = 0$. Notice that $h_t = 0$ for all t . Based on your answer to part (a), determine the value α such that if $w < \alpha$, the gradient vanishes, while if $w > \alpha$, the gradient explodes. You may use the fact that $\sigma'(0) = 1/4$. [8 pts]

6. Language Models and Transformers

Both

- (a) In the encoder/decoder architecture for machine translation, the network reads a sentence and stores all the information in its hidden units. Describe two ways in which a bidirectional RNN encoder differs from this. [8 pts]

bidirectional RNN encoder goes left $\rightarrow$ right
right $\rightarrow$ left, so it has info from the entire sentence at the same time (not just "past")
there is no decoder part looking at encoder representations

- (b) How do self-attention and cross-attention differ? [6 pts]

key/value

$K=V=Q$ vs $K=V$



(but Q is different; there is
in the decoder to ask for
noun or adj, and then V
 $K=V$ we find them in
sentence

In transformers, we
attention in encoder part
attention (Q from decoder,
encoder) in decoder part
(coming from the encoder)

- (c) Describe one way in which BERT and GPT models are similar, and one way in which they differ. [8 pts] encoder decoder

7. Autoencoders

(a) What is the training objective for a standard autoencoder? [5 pts]

minimize the difference between the input and the output (e.g. L_2 norm between x & $\hat{x}$, distance metrics can be used)

overall goal is to be able to compress the input into a useful lower dimensional representation & then reconstruct it

(b) How does the Variational Autoencoder differ from a standard autoencoder? [5 pts]

takes input x , maps it to latent space (lower dimensional). Takes a sample z from this latent space. Generates output $\hat{x}$ from z . It's stochastic (i.e. sampled from a distribution dependent on x could lead to different z samples) whereas standard autoencoder is not stochastic - we're not sampling anything.

(c) Describe the two terms in the training objective of a VAE, and how they relate to the overall aim of the model. [6 pts]